

chapters 7 and 10). The difference increases to 5.3% employing nonlinear, dynamic factors, which we consider below. These are large biases—*HF report returns that are too good to be true*.

Asset owners should also be wary of HF indexes. The average investor cannot obtain the returns reported in these indexes because HF indexes generally are *not investable*. There is no equivalent to a S&P 500 index fund that can cheaply replicate the aggregate stock market. Nor can asset owners invest in every HF in most HF indexes—some are closed, some have minimums that are too high, and in any case there are just too many of them. And when you try to get your money out, you can't (see below), making it impossible to bail when a fund has dropped out of the index. Since investors cannot hold all the funds in a HF index, they face more idiosyncratic risk investing in HFs than is measured by the HF index, which diversifies away much more risk than an investor can.²⁴ I use HF index returns in this chapter, but you've been warned: they generally give too rosy a picture of returns.

Since HFs voluntarily report returns, there are other biases. Agarwal, Daniel, and Naik (2011) find that Santa Claus is kind to HFs: returns are higher in December than between January and November by 1.3%, on average. HFs do this by underreporting their returns earlier in the year and by borrowing from January returns. Why do HFs inflate their December returns? Because HFs levy fees on their December numbers. Bollen and Pool (2009) show that there are more returns just above zero than below zero at the monthly frequency. (The zero level is especially important because HFs market themselves as absolute return strategies.) This kink disappears measuring returns in two-month intervals, indicating that some monthly returns are inflated to avoid reporting losses. These overstatements are subsequently reversed when HFs have stored up a bank of good returns.

Revisions of past data by HFs are pervasive. Patton, Ramadorai, and Streatfield (2013) claim that half of the 12,128 HFs in their sample over 2007–2011 revised their previously reported returns. One-fifth of HFs modified their returns by at least 1.0%—enormous revisions compared to the average monthly HF return of 0.6%. The best way (large) asset owners can avoid data revisions is to use *separately managed accounts*, where the asset owner directly holds the assets and the HF manager trades the securities in the account.²⁵ This doesn't help asset owners in analyzing past, reported data to ascertain whether they should invest in a HF in the first place. For that decision, the asset owner must realize that reported HF returns are too good to be true.

²⁴ Some firms have designed *investable* HF indexes, but there is a large selection bias in the types of HFs willing to be included in such indexes. As Groucho Marx says, "I refuse to join any club that would have me as a member."

²⁵ Aggarwal and Jorion (2012) find no evidence that more transparency is associated with lower returns. In fact, their results suggest the opposite: the more transparent the HF, the higher the returns.